

AWS LAMBDA

AWS Lambda is a serverless computing service in Amazon Web Services that lets you run code without managing servers. unused resources should be stopped or deleted to reduce cost and improve account efficiency.

Key Features

- Serverless (no server management)
- Pay only for execution time
- Auto scaling (runs many requests at once)
- Event-driven (runs only when triggered)
- Works with many AWS services

How it works

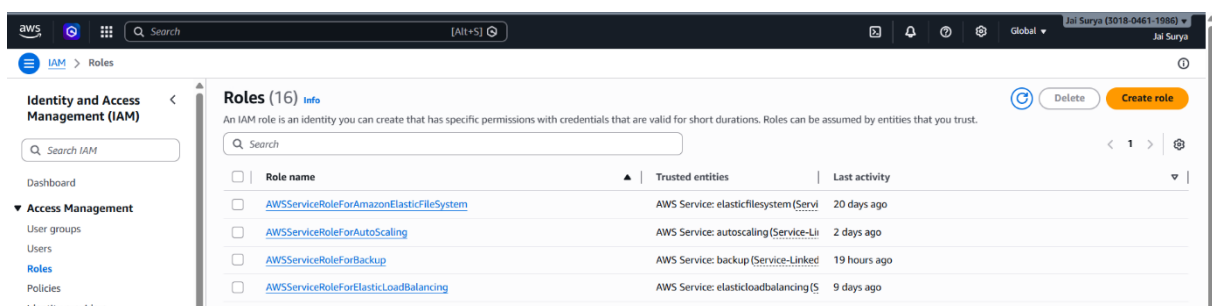
1. You upload your code (Python, Node.js, etc.)
2. AWS stores it in Lambda
3. An event triggers the function
 - Example: API request, file upload, database change
4. Lambda runs the code and returns output

HAND-ON FOR LAMBDA

DELETING/STOPING THE UNUSED INSTANCES

Screenshot: Create IAM Role

- Go to IAM
- Click Roles
- Click Create role



Screenshot: Select Trusted Entity

- Choose AWS service
- Select Lambda under use case

The screenshot shows the 'Select trusted entity' step in the AWS IAM console. The left sidebar indicates the current step is 'Step 1: Select trusted entity'. The main content area is titled 'Select trusted entity' and contains two sections: 'Trusted entity type' and 'Use case'.

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a third party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
Lambda

Choose a use case for the specified service.

Use case

- ☒ **Lambda**
Allows Lambda functions to call AWS services on your behalf.
- ☐ **AWSServiceRoleForLambda**
Allows Lambda to manage AWS resources on your behalf.

Buttons: Cancel, Next

Screenshot: Attach Permissions

- Search for EC2 Full Access
- Select the policy

The screenshot shows the 'Add permissions' step in the AWS IAM console. The left sidebar indicates the current step is 'Step 2: Add permissions'. The main content area is titled 'Add permissions' and contains a search bar, a filter dropdown, and a list of permissions policies.

Permissions policies (1/1136)

Choose one or more policies to attach to your new role.

Filter by Type: All types (36 matches)

Search: ec2

Policy name	Type	Description
☐ AmazonEC2ContainerRegistryFullAccess	AWS managed	Provides administrative access to Amazon EC2 Container Registry.
☐ AmazonEC2ContainerRegistryPowerUser	AWS managed	Provides full access to Amazon EC2 Container Registry.
☐ AmazonEC2ContainerRegistryPullOnly	AWS managed	Provides access to pull images from Amazon EC2 Container Registry.
☐ AmazonEC2ContainerRegistryReadOnly	AWS managed	Provides read-only access to Amazon EC2 Container Registry.
☐ AmazonEC2ContainerServiceAutoscaleRole	AWS managed	Policy to enable Amazon EC2 Container Service to use Amazon EC2 Autoscaling.
☐ AmazonEC2ContainerServiceEventsRole	AWS managed	Policy to enable Amazon EC2 Container Service to use Amazon CloudWatch Events.
☐ AmazonEC2ContainerServiceforEC2Role	AWS managed	Default policy for Amazon EC2 Container Service for Amazon EC2.
☐ AmazonEC2ContainerServiceRole	AWS managed	Default policy for Amazon EC2 Container Service.
☒ AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2.
☐ AmazonEC2ImageReferencesAccessPolicy	AWS managed	Provides read-only access to Amazon EC2 Image References.

Screenshot: Add Role Name

- Enter role name
- Click Create role
- IAM role created successfully

IAM > Roles > Create role

Step 1: Select trusted entity
Step 2: Add permissions
Step 3: **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

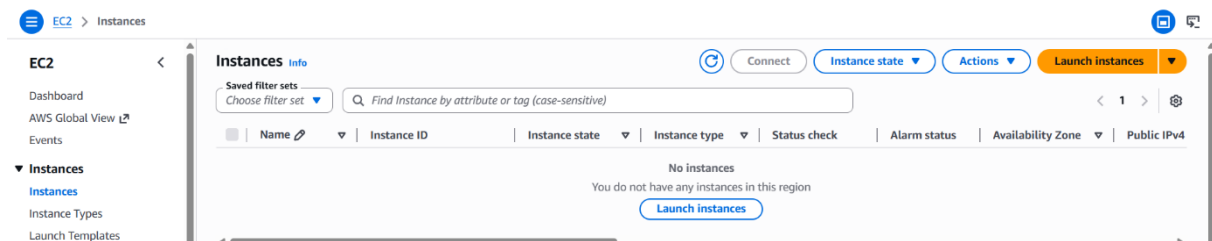
Maximum 64 characters. Use alphanumeric and "+,=, @, _" characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: "_+=, @-/\[\]\#%*&~;:'\"/>

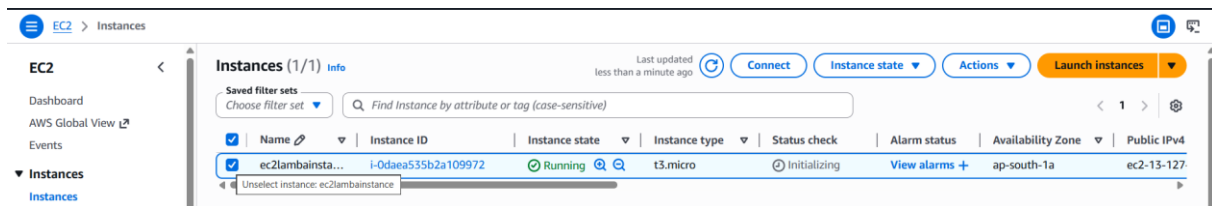
Screenshot: Open EC2

- Go to EC2 service
- Click Instances
- Click Create instance



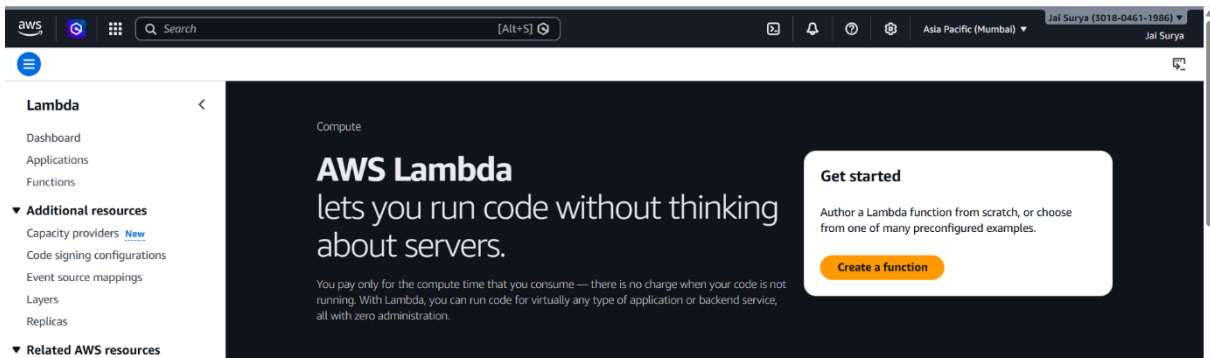
Screenshot: Instance Created

- EC2 instance created successfully
- Instance is in running state



Screenshot: Open AWS Lambda

- Go to AWS Console
- Search Lambda
- Click Lambda service



Screenshot: Create Lambda Function

- Select Author from scratch
- Enter function name
- Choose Python runtime
- Click Create function

Lambda > Functions > Create function

Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

mylambda

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime | [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Durable execution - new | [Info](#)
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#)

☐ Enable

Screenshot: Configure Execution Role

- In Execution role select Use an existing role
- Choose the created IAM role
- Click Create function

▼ Change default execution role

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role
 ☒ Use another role

ec2lambda

▼

⌄

Create new role

[View role details in IAM](#)

▶ Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

Cancel

Create function

Screenshot: Lambda Function Created

- Lambda function created successfully
- Function is now ready for configuration and testing

aws

⌄

Search

[Alt+S]

📧

🔔

?

⚙️

Asia Pacific (Mur

≡

[Lambda](#) > [Functions](#) > mylambda

⌄

Successfully created the function mylambda. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

×

mylambda

Throttle

Copy ARN

Actions ▼

Export to Infrastructure Composer

Download ▼

▼ Function overview

Info

Diagram

Template

mylambda

Layers (0)

+ Add trigger

+ Add destination

Function ARN

arn:aws:lambda:ap-south-1:301804611986:func:mylambda

Description

-

Last modified

11 seconds ago

Code

Test

Monitor

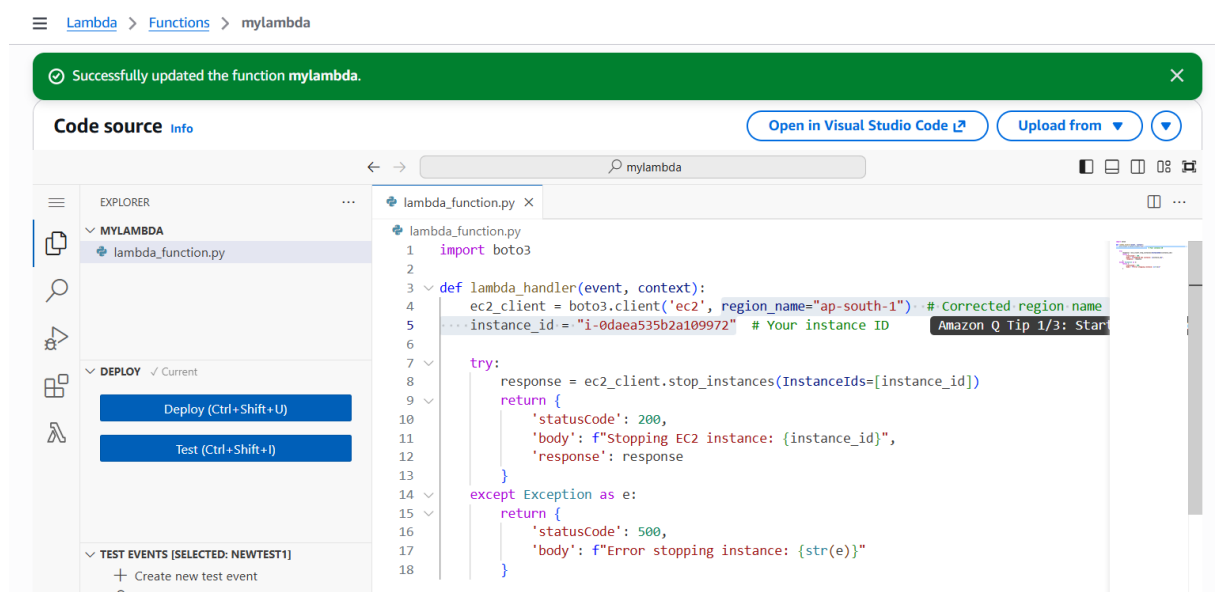
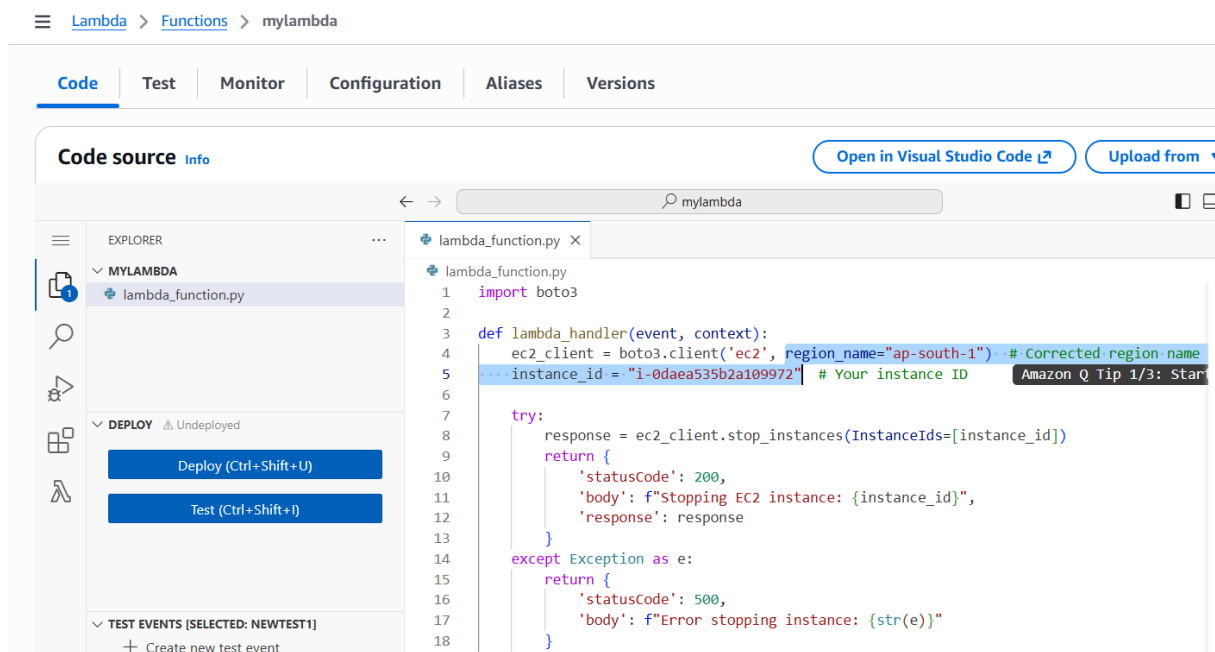
Configuration

Aliases

Versions

Screenshot: Add Lambda Code

- Go to Code section
- Paste code to stop unused EC2 instance
- Add EC2 instance ID in the code
- Add region/location in the code
- Click Deploy



Screenshot: Configure General Settings

- Go to Configuration tab
- Click Edit in General configuration

CodeTestMonitorConfigurationAliasesVersions

Configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

General configuration

Description

-

Memory

128 MB

Ephemeral storage

512 MB

Timeout

0 min 3 sec

SnapStart

None

Edit

Screenshot: Set Timeout

- Set timeout value
- Click Save

SnapStart

Info

Reduce startup time by having Lambda cache a snapshot of your function after the function has initialized. To evaluate whether your function code is resilient to snapshot operations, review the [SnapStart compatibility considerations](#). For Python and .NET runtimes, view [pricing](#).

None

Supported runtimes: .NET 10 (C#/F#/PowerShell), .NET 8 (C#/F#/PowerShell), Java 11, Java 17, Java 21, Java 25, Python 3.12, Python 3.13, Python 3.14.

Timeout

5 min 0 sec

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

ec2lambda

Create new role

[View role details in IAM](#)

Cancel

Save

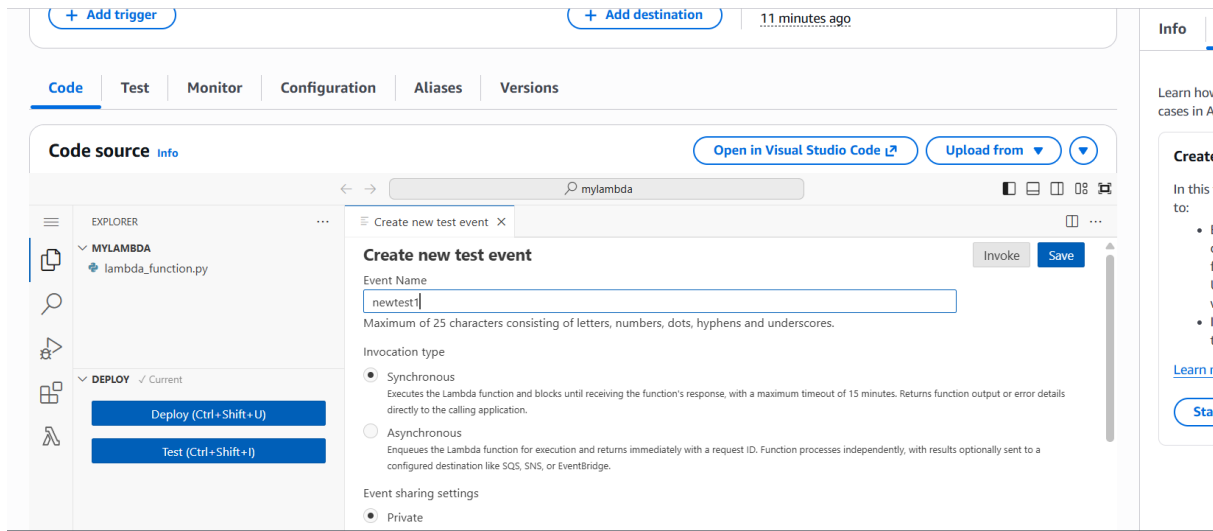
Lambda > Functions > mylambda

Successfully updated the function mylambda.

Ir

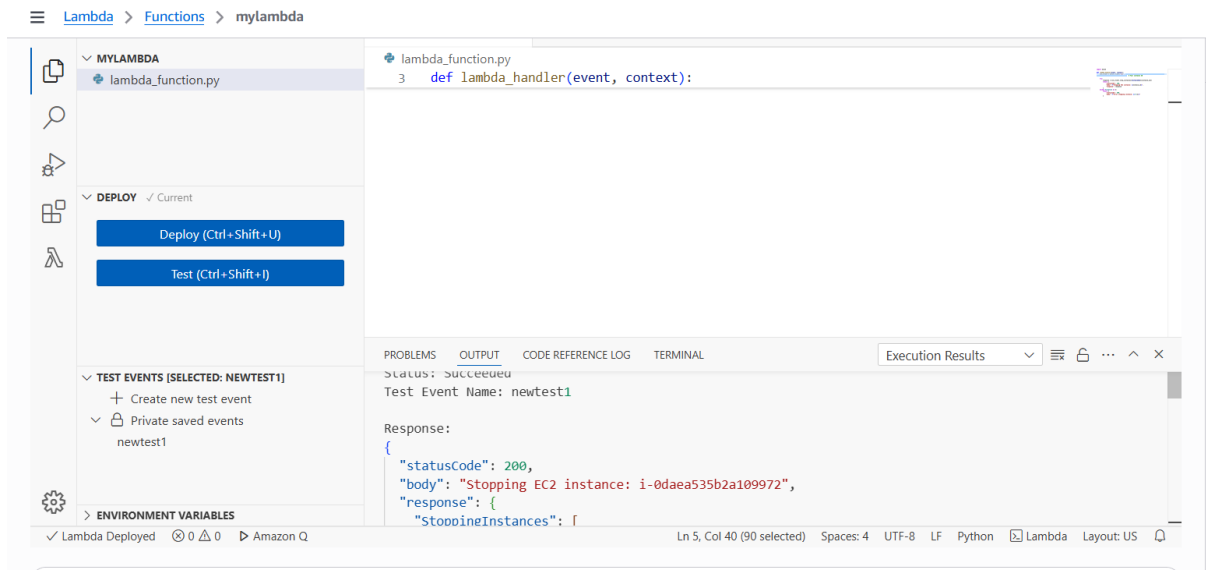
Screenshot: Create Test Event

- Go to Code section
- Click Test
- Enter test event name
- Click Create test event



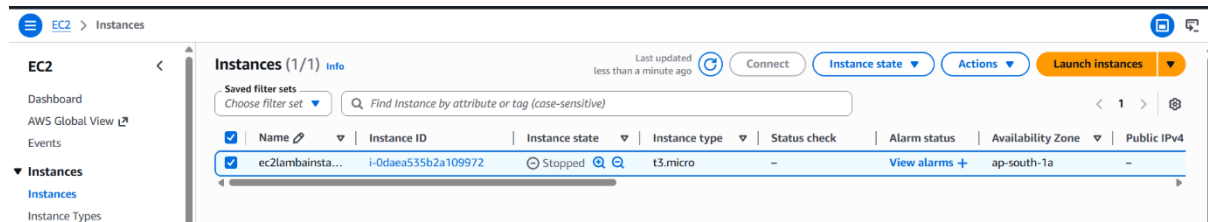
Screenshot: Run Test

- Click Test
- Execution starts
- Output shows function is running successfully



Screenshot: Verify EC2 Instance Status

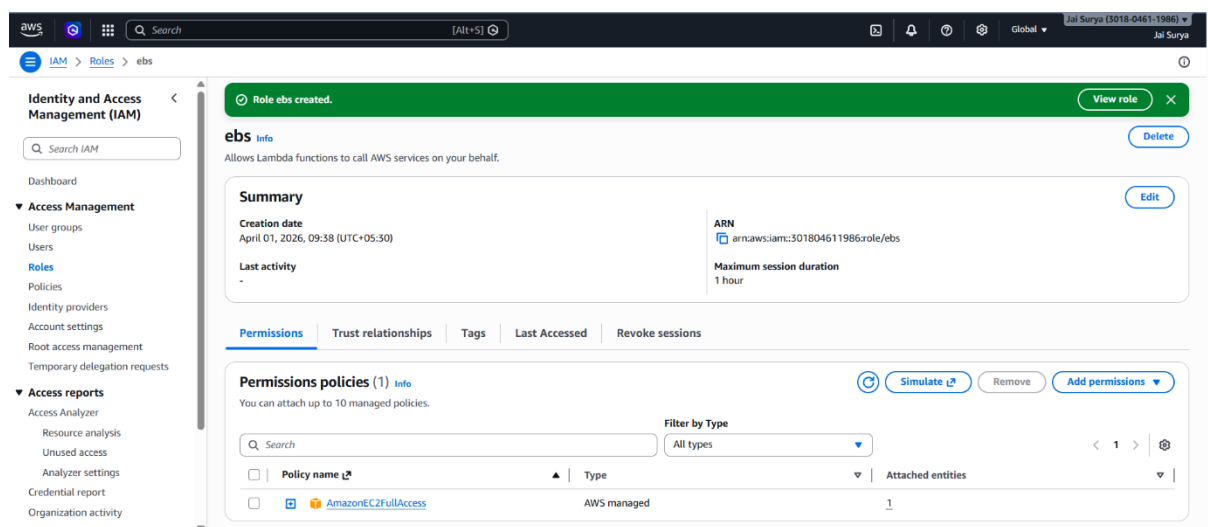
- Go to EC2 service
- Select the instance
- Check instance status is Stopped



DELETING UNUSED EBS VOLUMES USING LAMBDA

Screenshot: Create IAM Role, EBS Volume, Lambda Function and Execute Deletion Task

- Go to IAM service and click Roles
- Click Create role and select AWS service
- Choose Lambda use case and attach EC2 Full Access permission
- Enter role name and create role
- Go to EC2 service and click Volumes
- Click Create volume and configure size/type
- Go to Lambda service and click Create function
- Select Author from scratch, enter function name, choose Python runtime
- Select existing IAM role (created EBS role) and create function
- Go to Code section, paste EBS volume deletion code and click Deploy
- Click Test and create test event
- Run the test and check output showing successful execution
- Go to EC2 Volumes and verify volume is deleted successfully



Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Volumes (1) info

Last updated less than a minute ago

Recycle Bin

Actions

Create volume

Saved filter sets

Choose filter set

Search

< 1 >

Table with 10 columns: Name, Volume ID, Type, Size, IOPS, Throughput, Snapshot ID, Source volume ID, Created

Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Successfully created volume vol-0d80d3cb53831d3e1.

Volumes (2/3) info

Last updated less than a minute ago

Recycle Bin

Actions

Create volume

Saved filter sets

Choose filter set

Search

< 1 >

Table with 10 columns: Name, Volume ID, Type, Size, IOPS, Throughput, Snapshot ID, Source volume ID, Created

Lambda > Functions > Create function

Create function info

Choose one of the following options to create your function.

Author from scratch

Use a blueprint

Container image

Basic information

Function name

Runtime

Change default execution role

Execution role

Create default role

Use another role

ebs

Create new role

Lambda > Functions > ebslambda

Successfully updated the function ebslambda.

EXPLORER

EBSLAMBDA

lambda_function.py

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [NONE SELECTED]

Create new test event

lambda_function.py

```
1 import boto3
2
3 ec2 = boto3.client('ec2')
4
5 def lambda_handler(event, context):
6     # Get all unused (available) volumes
7     volumes = ec2.describe_volumes(
8         Filters=[
9             {
10                 'Name': 'status',
11                 'Values': ['available']
12             }
13         ]
14     )
15
16     # Loop through volumes and delete
17     for volume in volumes['Volumes']:
18         volume_id = volume['VolumeId']
19         print(f"Deleting Volume: {volume_id}")
20         ec2.delete_volume(VolumeId=volume_id)
21
22     return "Unused EBS volumes deleted successfully"
```

Successfully updated the function ebslambda.

Timeout

5 min 0 sec

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

ebs

Create new role

View role details in IAM

Cancel Save

Code Test Monitor Configuration Aliases Versions

Code source Info

Open in Visual Studio Code Upload from

EXPLORER

EBSLAMBDA

lambda_function.py

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

lambda_function.py

```
1 import boto3
2
3 ec2 = boto3.client('ec2')
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5 def lambda_handler(event, context):
6     # Get all unused (available) volume
7     volumes = ec2.describe_volumes(
8         Filters=[
9             {
10                 'Name': 'status',
11                 'Values': ['available']
12             }
13         ]
14     )
15
16     # Loop through volumes and delete
```

Create new test event

Invoke Save

Event Name

testebbs

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

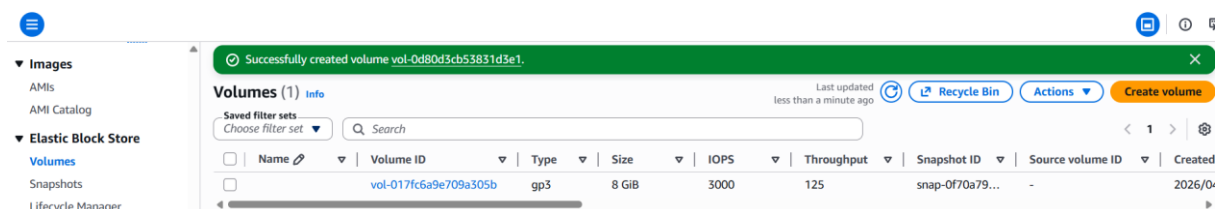
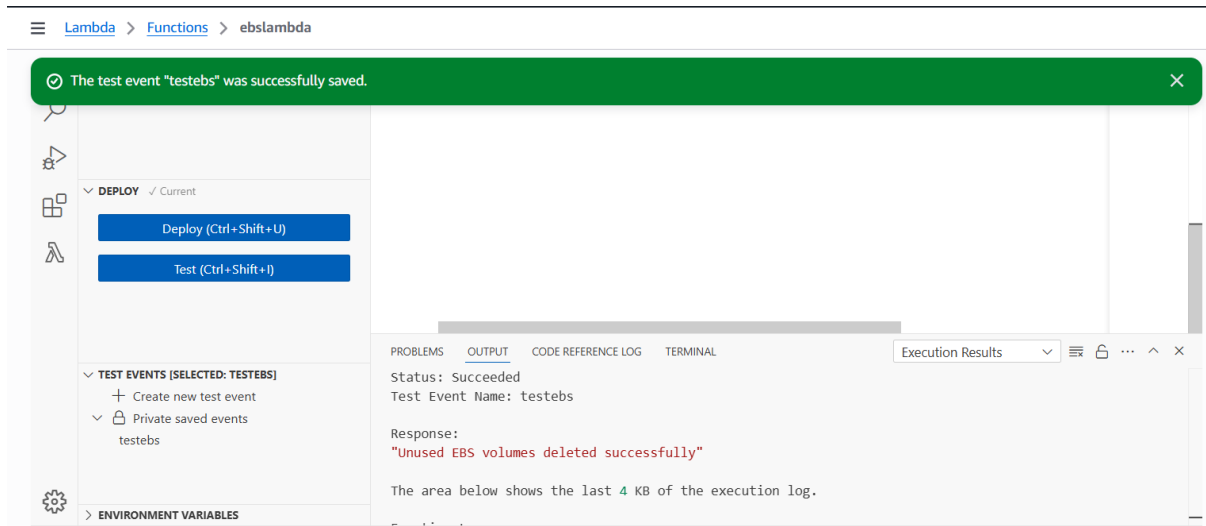
Invocation type

Synchronous

Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Asynchronous

Enqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.



DELETING UNUSED ELASTIC IPs USING LAMBDA

Screenshot: Delete Unused Elastic IP using Lambda

- Go to IAM service and click Roles
- Click Create role and select AWS service
- Choose Lambda use case and attach EC2 Full Access permission
- Enter role name and create role
- Go to EC2 service and allocate Elastic IP
- Go to Lambda service and click Create function
- Select Author from scratch, enter function name, choose Python runtime
- Select existing IAM role (created EC2 full access role) and create function
- Go to Code section, paste code to release unused Elastic IP and click Deploy
- Click Test and create test event
- Run the test to execute the function
- Check output showing Elastic IP released successfully
- Go to EC2 Elastic IP section and verify IP is released

EC2

Elastic IP addresses

Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Elastic IP addresses (1/1) Info

Find elastic IP addresses by attribute or tag

Public IPv4 address : 43.205.181.201

Clear filters

1

Actions

Allocate Elastic IP address

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associate
-	43.205.181.201	Public IP	eipalloc-0b56f487628a75cac	-	-

Lambda

Functions

Create function

Enter a name that describes the purpose of your function.

elasticip

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime Info

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

Durable execution - new Info

Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).

Enable

Architecture Info

Choose the instruction set architecture you want for your function code.

arm64

x86_64

Permissions Info

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

Change default execution role

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

Create default role

Use another role

ebs

Create new role

Lambda

Functions

elasticip

Successfully created the function elasticip. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

Code

Test

Monitor

Configuration

Aliases

Versions

Code source Info

Open in Visual Studio Code

Upload from

elasticip

lambda_function.py

lambda_function.py

def lambda_handler(event, context):

Get all Elastic IPs

addresses = ec2.describe_addresses()

Loop through and release unused EIPs

for addr in addresses['Addresses']:

if 'InstanceId' not in addr:

allocation_id = addr['AllocationId']

print(f"Releasing Elastic IP: {allocation_id}")

ec2.release_address(AllocationId=allocation_id)

return "Unused Elastic IPs released successfully"

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

SnapStart

Info

Reduce startup time by having Lambda cache a snapshot of your function after the function has initialized. To evaluate whether your function code is resilient to snapshot operations, review the [SnapStart compatibility considerations](#). For Python and .NET runtimes, [view pricing](#).

None

▼

Supported runtimes: .NET 10 (C#/.NET Core), .NET 8 (C#/.NET Core), Java 11, Java 17, Java 21, Java 25, Python 3.12, Python 3.13, Python 3.14.

Timeout

5 min 0 sec

Execution role

To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

ebs

▼

⌂

Create new role

[View role details in IAM](#)

Cancel

Save

Code source

Info

Open in Visual Studio Code

Upload from

▼

▼

EXPLORER

▼ ELASTICIP

lambda_function.py

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

lambda_function.py

5 def lambda_handler(event, context):

6

7

8 # Loop through and release unused EIPs

9

10 for addr in addresses['Addresses']:

11 if 'InstanceId' not in addr:

12 allocation_id = addr['AllocationId']

13 print(f'Releasing Elastic IP: {alloc

14

15 ec2.release_address(AllocationId=all

16

17 return "Unused Elastic IPs released successf

Create new test event

Invoke

Save

Create new test event

Event Name

eip

Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Invocation type

Synchronous

Executes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

Asynchronous

Enqueues the Lambda function for execution and returns immediately with a request ID.

Lambda > Functions > elasticip

Successfully updated the function elasticip.

The test event "eip" was successfully saved.

DEPLOY

Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [SELECTED: EIP]

+ Create new test event

Private saved events

eip

ENVIRONMENT VARIABLES

Lambda Deployed

0

Amazon Q

PROBLEMS

OUTPUT

CODE REFERENCE LOG

TERMINAL

Status: Succeeded

Test Event Name: eip

Response:

"Unused Elastic IPs released successfully"

The area below shows the last 4 KB of the execution log.

Execution Results

Test event is saved successfully.

EC2 > Elastic IP addresses

Elastic IP addresses

Info

Find elastic IP addresses by attribute or tag

Public IPv4 address: 43.205.181.201

Clear filters

Name

Allocated IPv4 addr...

Type

Allocation ID

Reverse DNS record

Associated instance ID

No matching resource found

Allocate Elastic IP address

AWS ROUTE 53

Amazon Route 53 is a scalable Domain Name System (DNS) web service in AWS used to route users to applications using domain names.

Route 53 converts domain names into IP addresses and routes traffic to AWS resources.”

Uses of Route 53

- Domain registration (buy/manage domains)
- DNS management (map domain → IP address)
- Route traffic to EC2, Load Balancer, S3, etc.
- Health checking of resources
- Failover routing for high availability
- Load distribution across servers

Key Features

- Highly available and scalable DNS service
- Low latency routing
- Domain registration support
- Health checks and monitoring
- Traffic routing policies
- Integration with AWS services

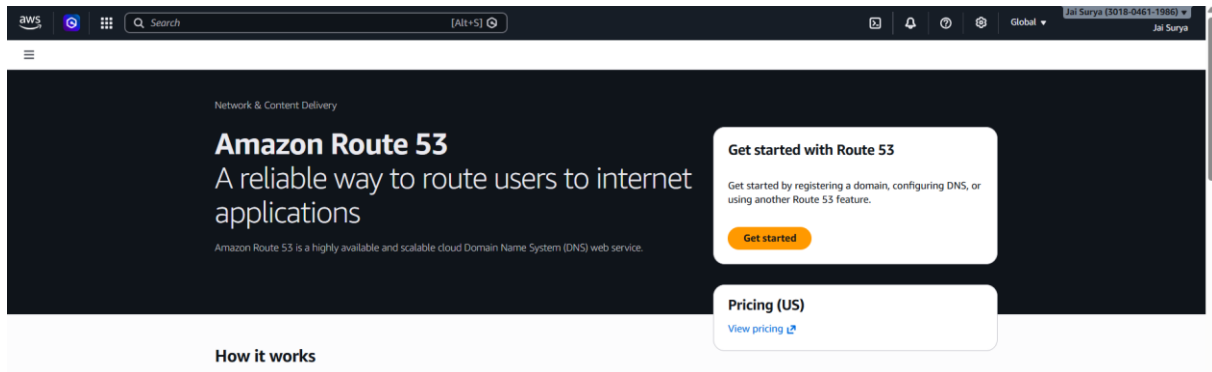
Route 53 Routing Policies

- **Simple Routing** – Routes traffic to a single resource (one IP or server).
- **Weighted Routing** – Distributes traffic based on assigned weight (e.g., 70% / 30%).
- **Latency Routing** – Sends users to the region with lowest latency (fastest response).
- **Failover Routing** – Routes traffic to a backup resource if primary fails.
- **Geolocation Routing** – Routes traffic based on user’s location (country/region).
- **Multi-Value Routing** – Returns multiple healthy IP addresses for better availability.
- **Geoproximity Routing** – Routes traffic based on closeness to AWS resources, with bias control.
- **IP-Based Routing** – Routes traffic based on the user’s IP address range (CIDR blocks).

HANDS-ON FOR ROUTE 53

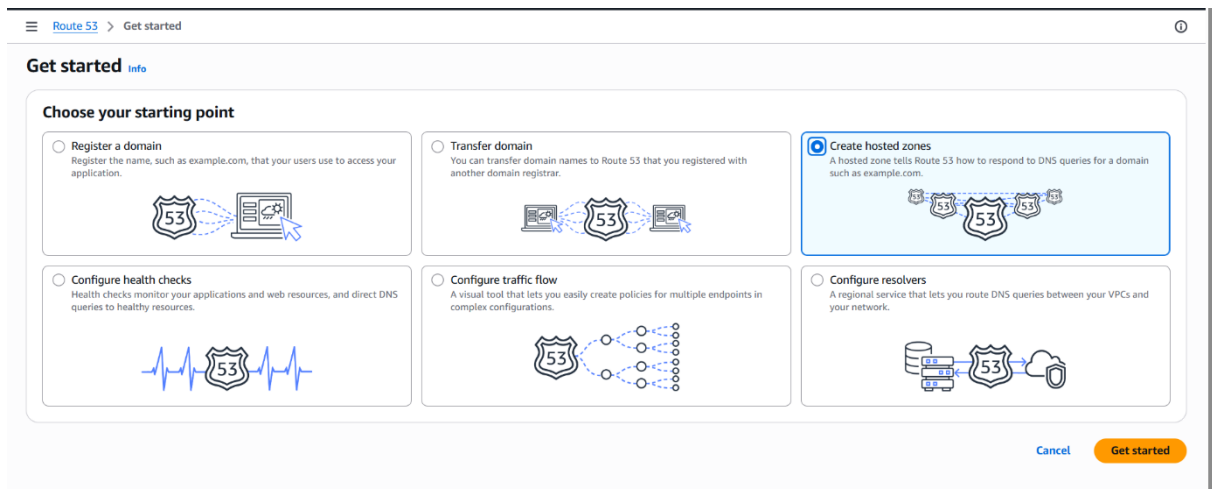
Screenshot: Open Route 53

- Go to AWS Console
- Search Route 53
- Click Route 53 service
- Click Get started



Screenshot: Create Hosted Zone

- Click Create hosted zone



Screenshot: Configure Hosted Zone

- Enter domain name
- Select Public Hosted Zone
- Click Create hosted zone

Route 53 > Hosted zones > Create hosted zone

Create hosted zone [Info](#)

Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.

Domain name [Info](#)
This is the name of the domain that you want to route traffic for.

 Valid characters: a-z, 0-9, ! * # \$ % & ' () * + , - / : ; < = > ? @ [\] ^ _ ` { | } . ~

Description - optional [Info](#)
This value lets you distinguish hosted zones that have the same name.

 The description can have up to 256 characters. 0/256

Type [Info](#)
The type indicates whether you want to route traffic on the internet or in an Amazon VPC.

☒ **Public hosted zone**
 A public hosted zone determines how traffic is routed on the internet.

☐ **Private hosted zone**
 A private hosted zone determines how traffic is routed within an Amazon VPC.

Screenshot: Hosted Zone Created

- Hosted zone created successfully
- Click Create record

Route 53 > Hosted zones > jai.com

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles
- ▼ **Global Resolver**
 - Global resolvers [New](#)
- ▼ **VPC Resolver**
 - VPCs
 - Inbound endpoints
 - Outbound endpoints
 - Rules
 - Query logging
 - Outposts
- ▼ **Domains**
 - Registered domains
 - Requests
- ▼ **IP-based routing**
 - CIDR collections
- ▼ **Traffic flow**

jai.com [Info](#) [Delete zone](#) [Test record](#) [Configure query logging](#)

Public

Hosted zone details [Edit hosted zone](#)

Records (2) Accelerated recovery DNSSEC signing Hosted zone tags (0)

Records (2) [Info](#) [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 > ⚙

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...)
☐	jai.com	NS	Simple	-	No	ns-1710.awsdns-21.co.uk. ns-413.awsdns-51.com. ns-969.awsdns-57.net. ns-1391.awsdns-45.org.	172800
☐	jai.com	SOA	Simple	-	No	ns-1710.awsdns-21.co.uk. a...	900

0 records selected

Select a record to see its details

Screenshot: Create Simple Routing Record

Sends traffic to **one single resource**

- Example: one EC2 instance for a website

The screenshot shows the 'Create record' page in the AWS Route 53 console. The breadcrumb trail is 'Route 53 > Hosted zones > jai.com > Create record'. The page title is 'Create record' with an 'Info' link. Under 'Quick create record', there is a 'Record 1' section. The 'Record name' is 'surya.com' and the 'Record type' is 'A - Routes traffic to an IPv4 address and some AWS resources'. The 'Value' is '12.0.0.0.6'. The 'TTL (seconds)' is '300'. The 'Routing policy' is 'Simple routing'. There are buttons for 'Add another record', 'Cancel', and 'Create records'.

Screenshot: Create Weighted Routing Record

Traffic is distributed based on weight

- Example:
 - EC2-A → 70% traffic
 - EC2-B → 30% traffic

Used for testing and load balancing

This screenshot shows the 'Create record' page for a weighted routing record. The 'Record name' is 'surya.com' and the 'Record type' is 'A - Routes traffic to an IPv4 address and some AWS resources'. The 'Value' is '12.0.0.0.6'. The 'TTL (seconds)' is '300'. The 'Routing policy' is 'Weighted'. The 'Weight' is '128'. There is a 'Health check ID - optional' field with a search icon. The 'Record ID' is 'rec-1'. Buttons for 'Add another record', 'Cancel', and 'Create records' are at the bottom.

Screenshot: Create Geolocation Routing Record

Routes traffic based on user location (country/region)

- Example:
 - India users → India server
 - US users → US server

Route 53 > Hosted zones > jai.com > Create record

Record name [Info](#)
 jai.com
Keep blank to create a record for the root domain.

Record type [Info](#)
A - Routes traffic to an IPv4 address and some AWS resources

☒ Alias

Value [Info](#)

Enter multiple values on separate lines.

TTL (seconds) [Info](#)
 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)
Geolocation

Location
Choose a location that DNS queries originate from. Queries from that location are routed to the resource that you specify in this record. Choose Default to route queries from locations that you don't create records for.
Europe

Health check ID - optional [Info](#)

Record ID [Info](#)

Add another record

Cancel Create records

View existing records

Screenshot: Create Latency Routing Record

Sends users to the lowest latency region

- Example:
 - India user → Mumbai region
 - US user → N. Virginia region

Route 53 > Hosted zones > jai.com > Create record

Record name [Info](#)
 jai.com
Keep blank to create a record for the root domain.

Record type [Info](#)
A - Routes traffic to an IPv4 address and some AWS resources

☒ Alias

Value [Info](#)

Enter multiple values on separate lines.

TTL (seconds) [Info](#)
 1m 1h 1d
Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)
Latency

Region
The Amazon EC2 region where the resource that you specified in this record resides. You can only create one latency record for each Amazon EC2 region. You aren't required to create latency records for all Amazon EC2 regions.
Asia Pacific (Mumbai)

Health check ID - optional [Info](#)

Record ID [Info](#)

Add another record

Cancel Create records

Screenshot: Create Health Check for Failover Routing

Route 53 > Health checks

Route 53
Dashboard
Hosted zones
Health checks
Profiles

Global Resolver
Global resolvers [New](#)

VPC Resolver
VPCs
Inbound endpoints
Outbound endpoints

Health checks (0) [Info](#)
Route 53 health checks monitor the health and performance of your application's servers and endpoints.

ID	Name	Details	Status in last 24 hours	Current sta...	Alarm	State	Actions
No health checks to display.							

Create health check

Route 53 > Health checks > Create health check

Create health check Info

Create a new health check to monitor the health of a specified resource.

Configuration

Configuration options have cost implications. Pricing will vary based on the selected resource type. Advanced configurations have an additional cost. [View pricing](#).

Name - optional
A friendly name that lets you easily find a health check in the dashboard.

The name must have 1-256 characters. Valid characters: A-Z, a-z, 0-9, -, (hyphen), and _ (underscore).

Resource

☒ **Endpoint**
Establish a connection with the resource to determine its health status.

☐ **Calculated health check**
The status of the health check is based on the status of the other health checks.

☐ **CloudWatch alarm**
The status of the health check is based on the state of a specified CloudWatch alarm.

Specify endpoint by

☒ **IP address**

☐ **Domain name**

IP address
The path can be any value for which your endpoint will return an HTTP status code of 2xx or 3xx when the endpoint is healthy.

Advanced configuration

Tags - optional

A tag is a label that you assign to an AWS resource. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

[Add new tags](#)

You can add up to 50 more tags.

[Cancel](#)
[Create health check](#)

aws [Search] [Alt+S]

Route 53 > Health checks > qqqq

Route 53

- Dashboard
- Hosted zones
- Health checks
- Profiles
- ▼ Global Resolver
 - Global resolvers [New](#)
- ▼ VPC Resolver

Health check with id c81e6fa2-cb56-45d0-b21d-d9d171023bfe has been created successfully.

[Delete](#) [Invert](#) [Disable](#) [Edit](#)

qqqq Info

Configuration	URL http://12.0.0.3:80/	Specified endpoint by IP address
ID c81e6fa2-cb56-45d0-b21d-d9d171023bfe	Status Unknown	Inverted No
State Enabled		

Screenshot: Create Failover Routing Record

Used for backup system

- Primary server active
- If fails → traffic goes to secondary

Route 53 > Hosted zones > jai.com > Create record

Create record Info

Quick create record

▼ Record 1

Record name Info

 jai.com

Record type Info

A - Routes traffic to an IPv4 address and some AWS resources

Keep blank to create a record for the root domain.

☒ **Alias**

Value Info

Enter multiple values on separate lines.

TTL (seconds) Info

Recommended values: 60 to 172800 (two days)

Routing policy Info

Failover

Failover record type

Choose **Primary** to route traffic to the specified resource by default or **Secondary** to route traffic to the specified resource when the primary resource is unavailable. You can create only one failover record of each type.

Primary

Health check ID Info

c81e6fa2-cb56-45d0-b21d-d9d171023bfe

Record ID Info

rec-1

Route 53 > Hosted zones > jai.com > Create record

Create record [Info](#)

[Switch to wizard](#) [Delete](#)

Record 1

Record name [Info](#) subdomain .jai.com
Keep blank to create a record for the root domain.

Record type [Info](#) A - Routes traffic to an IPv4 address and some AWS resources

☐ Alias

Value [Info](#) 12.9.9.9

Enter multiple values on separate lines.

TTL (seconds) [Info](#) 300 [1m](#) [1h](#) [1d](#) [Recommended values: 60 to 172800 \(two days\)](#)

Routing policy [Info](#) Failover

Fallover record type
Choose Primary to route traffic to the specified resource by default or Secondary to route traffic to the specified resource when the primary resource is unavailable. You can create only one fallover record of each type.

Secondary

Health check ID - optional [Info](#) Q. c81e6fa2-cb56-45d0-b21d-d9d171023bfe [X](#) [Refresh](#)

Record ID [Info](#) rec-1

Screenshot: Create Multi-Value Routing Record

Multi-Value Routing

Returns multiple healthy IPs

- Used for basic load balancing
- Improves availability

Route 53 > Hosted zones > jai.com > Create record

Create record [Info](#)

[Switch to wizard](#) [Delete](#)

Record 1

Record name [Info](#) surya.com .jai.com
Keep blank to create a record for the root domain.

Record type [Info](#) A - Routes traffic to an IPv4 address and some AWS resources

☐ Alias
You can't use an alias record with multivalue routing policy.

Value [Info](#) 12.9.9.9

Enter multiple values on separate lines.

TTL (seconds) [Info](#) 300 [1m](#) [1h](#) [1d](#) [Recommended values: 60 to 172800 \(two days\)](#)

Routing policy [Info](#) Multivalue answer

Health check ID - optional [Info](#) Q. c81e6fa2-cb56-45d0-b21d-d9d171023bfe [X](#) [Refresh](#)

Record ID [Info](#) rec-1

[Add another record](#)

[Cancel](#) [Create records](#)

Screenshot: Create CIDR Collection for IP-Based Routing

Route 53 > CIDR collections > Create CIDR collection

Create CIDR collection [Info](#)

To create a CIDR collection, you must create at least one CIDR location with CIDR blocks. All record sets with the same record set name and type must reference the same collection.

Details

Collection name

jai2003

Valid characters: a-z, 0-9, -, _

Create CIDR locations

To create a CIDR collection, you must create at least one CIDR location with CIDR blocks.

▼ CIDR location 1

CIDR location

ap-south-1

Valid characters: a-z, 0-9, -, _

CIDR blocks

List of IP addresses from /0 to /24 for IPv4 and /0 to /48 for IPv6.

192.0.0.0/16

[Add another location](#)

[Remove location](#)

[Add another location](#)

Screenshot: Create IP-Based Routing

Route 53 > Hosted zones > jai.com > Create record

Quick create record [Switch to wizard](#)

▼ Record 1

[Delete](#)

Record name [Info](#)

rec-1

jai.com

Keep blank to create a record for the root domain.

Record type [Info](#)

A - Routes traffic to an IPv4 address and some AWS resources

☒ **Alias**

Value [Info](#)

16.0.0.0

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300

1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

IP-based

IP-based

Locations that hold IP address ranges in CIDR notation.

ap-south-1

Health check ID - optional [Info](#)

Choose health check

Record ID [Info](#)

US West load balancer

[Add another record](#)

Screenshot: Create Geoproximity Routing Record

Route 53 > Hosted zones > jai.com > Create record

Create record [Info](#)

Quick create record

[Switch to wizard](#)

▼ Record 1

[Delete](#)

Record name [Info](#)

rec-1

jai.com

Keep blank to create a record for the root domain.

Record type [Info](#)

A - Routes traffic to an IPv4 address and some AWS resources

☒ **Alias**

Value [Info](#)

16.0.0.0

Enter multiple values on separate lines.

TTL (seconds) [Info](#)

300

1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy [Info](#)

Geoproximity

Endpoint location type

Coordinates

Longitude

9

Latitude

8

Bias

Bias is used to expand or shrink the size of the geographic area from which Route 53 routes traffic to a resource. A positive bias will expand whereas a negative bias will shrink. [Learn more about how bias affects geoproximity routing.](#)

4

Bias must be within the range of -99 and 99.

Health check ID - optional [Info](#)

Choose health check

Record ID [Info](#)

1